What a Working Life Holds: The Home, the Pension and the Portfolio in Hours of the World's Work

The household balance sheet measured in hours of work. Research note, v1.0, 21 October 2026. Keisuke Nakanishi.

This is a review-style note intended for open publication, not a journal submission. All series were built by the author from free public data: the Jordà–Schularick–Taylor Macrohistory Database, OECD analytical house price indicators and household balance sheets, the Federal Reserve's Financial Accounts of the United States (Z.1), Eurostat's national-accounts pension table, the Penn World Table, the World Bank, the UN World Population Prospects, BIS exchange rates and FRED, together with the frozen engine and the world hour behind the first three notes. No third-party proprietary data is redistributed, no index-vendor data is used, and no tables or figures from other publications are reproduced. Generative AI tools (Claude, Anthropic) were used for drafting, editorial review and writing analysis code, and as a discussion partner in analysis design. The author specified the questions, reviewed and ran all code, checked the outputs against the source data, and is responsible for all results. The analysis is exploratory: it was not pre-registered, and it was run with the results visible; the life-cycle design was corrected once after seeing results (Section 7). Sections 4–7 were each checked by a separate script written before reading the code that produced them. SSRN identifier: 7646578 (doi:10.2139/ssrn.7646578). Code and derived data: https://github.com/thecapitalcensus/working-life-holds.

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Summary

Note 3 asked what the global market portfolio (GMP) earns and pays in hours of the world's work (Nakanishi 2026d). This note asks the same of a household's four claims: home, pension, portfolio and deposits. **How many hours of work does each claim earn a year, and what does a working life end up holding?** It covers housing in 16 countries, 31 balance sheets and 476 stylised lives.

Five results follow.

1. **Before taxes and costs, the home earned about as much as the world's capital; after them, less.** Per 1,000 world hours, 1976–2025, housing gained a net 41 hours a year over the hour and the GMP 35; housing's surplus is its net rent. With a 1-point property tax and buying and selling every ten years, housing's 1976–2020 net falls from 40.0 to 22.0, against 38.0 for the GMP.
2. **Note 3's rent yields used wrong weights, now corrected.** Its v1.4 printed 5.5, 4.3 and 2.7% for 1976, 2000 and 2020; the corrected yields, now in note 3 v1.5, are 4.6, 4.5 and 3.7% (Section 4.4).
3. **Dwellings and land are the largest household asset; the pay-as-you-go pension claim is left off the balance sheet.** In 2023, dwellings and land were 41.6% of gross household assets (GDP-weighted). The accrued pay-as-you-go claim was a median 270% of GDP in 30 European countries in 2021, or 78% of household net worth where known (24 countries).
4. **The pension earns the growth of hours worked, or less.** A claim indexed to the wage bill earns, in hours, the growth of total hours worked: 6.6 per 1,000 a year in 1976–2020, negative in Germany, Japan and Finland. A claim indexed to average pay earns the growth of hours per worker, −3.6 per 1,000; over a life, a funded 60/40 claim ends 2.44 to 2.95 times larger.
5. **Over a life, buying a home and renting while investing come out about even.** Among 459 buyers, the owner's median lead at 65 is 0.08 years of final pay, and the owner wins in 51% of them. Across 162 variants of the assumptions that share ranges from 2% to 95%.

What a life holds depends on what each claim is indexed to. On one basis (national hours, 16 countries, 1976–2020), housing earned 3.96% a year over the hour and a wage-bill claim 0.44%. A claim indexed to work keeps pace with work by design, and little more.

1. The question

Note 3 asked three questions of a market: what a unit of capital generates in hours of work, what it pays out, and what it needs to stand still. This note asks the same questions of a household. A household does not hold the GMP alone. In most rich countries its largest asset is its home. Its largest claim is often a promise from the state to pay a pension that grows with the wage bill or with average pay, which does not appear on its balance sheet at all. Its portfolio and its deposits come after those.

Each of these claims has its own return in hours of work:

- *The portfolio* earns its total return less the growth of the hour. Note 3 measured this for the GMP.
- *The home* earns its net rent yield plus the growth of rents plus any change in the price paid per unit of rent, less the growth of the hour. Note 3 left this open.
- *A pay-as-you-go pension whose claims grow with the wage bill* — the return a balanced system can pay (Samuelson 1958; Aaron 1966) — earns the growth of the wage bill. Measured in hours, the growth of pay per hour cancels, and what is left is the growth of the number of hours worked: demography, participation and working time. A claim revalued with average pay per worker earns instead the growth of hours per worker.
- *Deposits and bonds* earn interest less the growth of the hour.

The pension point is the note's main conceptual step. A claim indexed to the wage bill is the one asset that is denominated in hours by construction. Its return in hours is not a market outcome; it is the change in how much work is done.

The note proceeds in four steps. Section 4 measures the home as note 3 measured the GMP. Section 5 sets out the household balance sheet. Section 6 measures the pension. Section 7 follows stylised lives from age 25 to 65 that hold all of these together.

2. Definitions

The world hour. The CC-WH series of Nakanishi (2026b): nominal GDP per hour worked, chained at GDP weights across 43 countries, in US dollars. It is hard data to June 2025 and provisional after. It is used as in note 3: December to December, anchored to the panel's dollars per hour in 2014 Q1.

The national hour. A country's nominal GDP in local currency divided by its total hours worked (World Bank GDP over Penn World Table employment × average annual hours, from 1961; Macrohistory Database GDP over Penn World Table hours for 1951–1960). A return "in national hours" is a local-currency return less the growth of this series. It answers a resident's question: does my claim command more of my own country's work?

Hours generated, needed and net. As in note 3: capital worth 1,000 hours that returns TR (log) generates 1,000 × TR hours' worth of currency in a year; 1,000 × g of those are needed to keep pace with an hour growing at g; the net gain is the difference. Every return in Sections 4–6 is in these units, hours a year per 1,000 hours of the claim. Ten hours per 1,000 is 1% a year (log).

Housing's parts. The total return on housing is the net rent yield (cash) plus the price change. The price change is split into a valuation change, minus the change in the log of the net rent yield, and the rest, "price excluding valuation", which is close to the growth of rents. Across countries, valuation is the GDP-weighted mean of each country's change in log rent yield, not the change in the yield of the 16-country portfolio as a whole (Section 4.1).

Years of own work, years of own pay. Net worth per worker measured in national hours, divided by annual hours per worker, equals net worth divided by GDP. The workers, the hours and the currency cancel. So "years of a country's own work" is simply the wealth-to-GDP ratio (checked to machine precision). In Section 7, "pay" is the national hour with hours per worker held fixed. Wealth divided by pay at 65 is therefore "years of own final pay", and equals wealth in national hours up to a constant. It is not years of measured earnings.

Two pension indexations. A claim *indexed to the wage bill* grows with pay per hour times total hours worked; this is the return a pay-as-you-go system can sustain. Notional defined-contribution schemes credit accounts with wage-bill, average-income or GDP growth, depending on the country. A claim *indexed to average pay* grows with pay per worker, as many national revaluation rules do. In national hours the first earns the growth of total hours worked, the second the growth of hours per worker.

3. Data and construction

Table 1. Data (all free and public; retrieved September–October 2026)

Source	Used for	Coverage	Terms
Jordà–Schularick–Taylor Macrohistory Database, release 6	House prices, housing returns and net rent yields; equity, bond and bill returns; long rates; GDP; exchange rates	16 countries with housing data, to 2020	CC BY-NC-SA 4.0
OECD analytical house price indicators	House price and rent indices chained onto 2020; cross-check	1970s–2025	OECD terms (reuse with attribution)
OECD household balance sheets: Table 720 (financial) and Table 9B (non-financial)	Composition of household assets and liabilities	31 countries with both sides in 2023	OECD terms
Federal Reserve, Financial Accounts of the United States (Z.1), table B.101	US household balance sheet	1952–2026 Q2	Public domain, attribution
Eurostat, pension entitlements in national accounts (Table 29: nasa_10_pens1)	Accrued pay-as-you-go and funded pension entitlements	30 countries, 2021	CC BY 4.0
Penn World Table 11.0 (also via FRED; the two copies are identical)	Persons engaged, average annual hours, labour share	1950–2023	CC BY 4.0
World Bank, World Development Indicators	GDP in local currency and US dollars; labour force (to extend hours to 2024–2025)	1960–2025	CC BY 4.0
UN World Population Prospects 2024	Population aged 20–64 (estimates and medium-variant projection)	1950–2100	CC BY 3.0 IGO
BIS exchange rates; FRED (BLS CPI rent of primary residence; US GDP; nonfarm business hours)	Exchange rates after 2020; US 2025 rent; US hours after 2023	to 2026	Public, attribution
Notes 1–3 and the GMP weight history (Nakanishi 2026a–d)	GMP total return and cash yield; the world hour	1976–2026	CC BY-NC 4.0 (CC-GMP TR, CC-WH)

Housing, to 2020. The Macrohistory Database, release 6 (Jordà, Schularick and Taylor 2017), gives annual housing total returns, capital gains and net rent yields for 16 countries (Jordà, Knoll, Kuvshinov, Schularick and Taylor 2019), built on the house price series of Knoll, Schularick and Steger (2017). Its net rent yield is rent less maintenance, management, insurance and depreciation, imputed for owners. The total return equals the capital gain plus the rent return to machine precision. The rent return equals this year's rent yield times one plus the capital gain, i.e. rent over the start-of-year price, to machine precision; Australia 1950 is the one exception. Countries are weighted by start-of-year US-dollar GDP. The US weight averages 41% over 1976–2020. Returns in US dollars use the database's exchange rate, `xrusd`, which is an end-of-year rate: Japan reads 200.5 for 1985 and 159.1 for 1986, the year-end rates, not the annual averages. December-to-December exchange rates therefore line up with the December-to-December world hour.

A unit problem in the database's GDP. The database's nominal GDP, `gdp`, is in millions, billions or trillions of local currency, depending on the country. Dividing it by the exchange rate does not give US dollars. Here each country's series is rescaled by a power of ten fitted to World Bank US-dollar GDP; the largest residual standard deviation, in $\log_{10}$, is 0.05. Note 3 did not do this (Section 4.4).

Housing, 2021–2025. The database ends in 2020. The OECD analytical house price indicators give a nominal house price index and a rent index for every one of the 16 countries. The price is chained onto the database's 2020 level with the OECD index (annual average). The rent yield is the database's 2020 yield times the growth of the rent index divided by the price index. Two gaps are filled: Australia's 2025 rent from the first three quarters, and US 2025 rent from the BLS consumer-price index for rent of primary residence. Exchange rates after 2020 are BIS December averages, within 2.3% of the database's 2020 rates. The OECD rent index measures rents paid by sitting tenants, not new leases; it lags market rents, so part of the 2021–2025 valuation change may be that lag.

Cross-check. Over 1976–2020, for each country, the database and the OECD indices agree closely: median correlations of annual growth are 0.98 for prices, 0.92 for the price–rent ratio and 0.93 for rents, and the median gap in price growth is +0.09 points a year. Substituting OECD rent growth for the database's implied rent growth (714 country-years) moves the 1976–2020 world-hour net from 40.0 to 41.7 (Table A5).

Balance sheets. OECD Table 720 (financial assets and liabilities) and Table 9B (non-financial assets), for households and the non-profit institutions serving them, in national currency at year-end. Gross assets are financial assets plus all non-financial assets: dwellings, all land held by households, and other produced assets. "Dwellings + land" therefore includes all household land, not only the land under homes. The latest year with at least 90% of the full country count is 2023. The US balance sheet from 1952 is the Federal Reserve's Z.1 table B.101; OECD US net worth is 1.07 times the Z.1 figure in 2023.

Pension entitlements. Eurostat's Table 29 records the accrued-to-date value of pension promises, including the social-security schemes that the core accounts leave out, every three years (Eurostat and ECB 2020). These are present values under each country's own discount rate and indexation assumptions, so levels differ for reasons other than the promises themselves.

Hours worked and population. Penn World Table 11.0 (Feenstra, Inklaar and Timmer 2015) gives persons engaged and average annual hours to 2023; their product is total hours. Sections 4 and 7 read them through FRED's copies, which are identical to the PWT 11.0 file used in Section 6. For 2024–2025 (housing's national hour only), total hours are extended with World Bank labour-force growth. The UN World Population Prospects 2024 gives the population aged 20–64, as estimates to 2023 and as the medium-variant projection after.

From note 3. The GMP's monthly total return and cash yield, built on the published weight history (Nakanishi 2026c), and the world hour, are read unchanged from the earlier notes' frozen files, whose checksums are listed in the Appendix.

4. The home in hours

4.1 Per 1,000 hours of housing capital

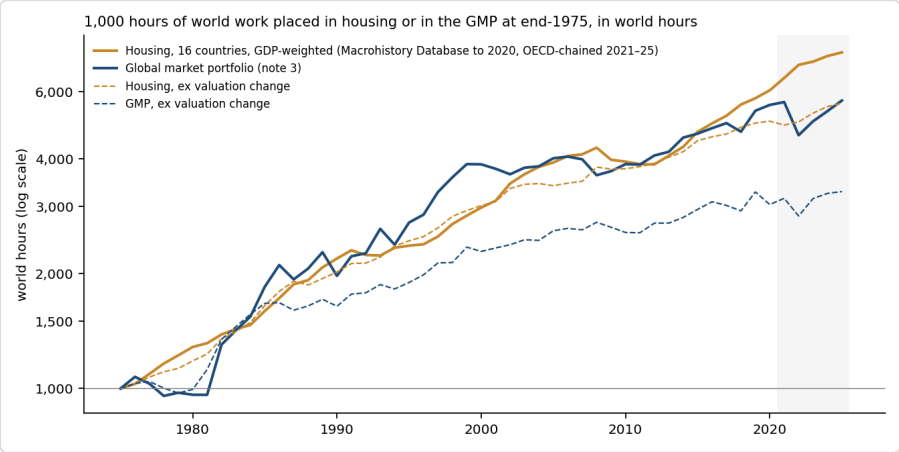


Figure 1. 1,000 hours of world work placed in housing (16 countries, GDP-weighted) or in the GMP at the end of 1975, in world hours, with and without valuation changes. Log scale. Shaded: 2021–2025, chained with OECD indices. Author's calculations.

Table 2. Hours a year generated by housing and by the GMP, per 1,000 world hours of capital (author's calculations). Annual log returns, % a year × 10. Housing: Macrohistory Database to 2020, OECD-chained 2021–2025, GDP-weighted at start-of-year US-dollar GDP. Components may not add exactly because of rounding

	1976–1981	1982–2000	2001–2020	2021–2025	1976–2020	1976–2025
Housing, 16 countries						
Total return (a)	141.0	85.9	77.5	81.9	89.5	88.8
— net rent / cash	46.7	45.9	41.4	34.2	44.0	43.0
— price	94.3	40.0	36.0	47.8	45.5	45.7
—— valuation change	10.8	−4.0	9.9	24.4	4.1	6.2
—— price excluding valuation	83.5	44.1	26.2	23.4	41.4	39.6
Needed to keep pace with the hour (c)	95.4	42.8	42.1	36.1	49.5	48.2
Net gain: (a) − (c)	**45.6**	**43.1**	**35.4**	**45.8**	**40.0**	**40.6**
Net gain excluding valuation	34.7	47.1	25.5	21.5	35.9	34.4
GMP (note 3)						
Total return (a)	89.0	116.1	60.0	41.4	87.6	83.0
— cash	78.9	64.5	33.9	29.2	52.8	50.4
— price	10.2	51.6	26.1	12.3	34.8	32.5
—— valuation change	−25.5	35.7	3.8	−10.3	13.4	11.0
—— price excluding valuation	35.6	15.9	22.3	22.6	21.4	21.5
Needed to keep pace with the hour (c)	95.4	42.8	42.1	36.1	49.5	48.2
Net gain: (a) − (c)	**−6.4**	**73.3**	**17.9**	**5.3**	**38.0**	**34.8**
Net gain excluding valuation	19.1	37.5	14.1	15.7	24.7	23.8

Over 1976–2020, housing generated 89.5 hours a year per 1,000 hours of housing capital, against 87.6 for the GMP. The two totals are close. Their make-up is not. Housing paid 44.0 hours as net rent; the GMP paid 52.8 in cash. Valuation added 4.1 hours to housing and 13.4 to the GMP. Both needed 49.5 hours to keep pace with the hour. So housing gained a net 40.0 hours a year and the GMP 38.0; excluding valuation, 35.9 and 24.7. Extended to 2025, the nets are 40.6 and 34.8, and 34.4 and 23.8 excluding valuation.

How valuation is measured matters a little. Valuation here is the GDP-weighted mean of each country's change in log rent yield. Measured instead as the change in the yield of the 16-country portfolio as a whole (same weights in both years), it is 3.9 hours rather than 4.1; the 0.25-hour difference sits inside "price excluding valuation", and net excluding valuation becomes 36.1 instead of 35.9. Price excluding valuation is close to, but not equal to, the growth of rents: $\ln(1 + g + y) - \ln(1 + y) \neq \ln(1 + g)$, where g is the capital gain and y the rent return. The mean gap over 1976–2020 is 0.24 points a year, and the largest is 1.8.

The periods differ more for the GMP than for housing. In 1976–1981, housing gained 45.6 hours a year net while the GMP lost 6.4. In 1982–2000 the GMP gained 73.3 against housing's 43.1. In 2001–2020 housing led again, 35.4 to 17.9. In 2021–2025 housing gained 45.8 and the GMP 5.3, but 24.4 of housing's hours were valuation: house prices outran rents. In the 26 twenty-year windows within 1976–2020, housing beat the GMP in 13. Housing's net ranged from 26 to 48 hours a year; the GMP's from 14 to 70.

Over 1976–2020, the annual net of housing had a standard deviation of 3.3%, against 8.2% for the GMP, and the two were uncorrelated (−0.05). Much of that smoothness belongs to the index, not to a house. House-price indices are often annual averages and appraisal-based, and an index of many houses has none of the risk of one house.

4.2 In each country's own hours

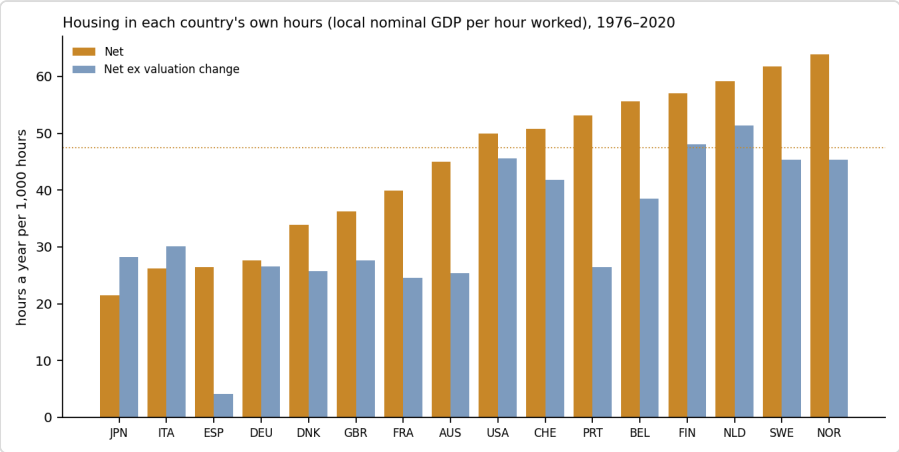


Figure 2. Net hours a year per 1,000 hours of housing capital, in each country's own hours (local nominal GDP per hour worked), 1976–2020, with and without valuation change. Dotted line: country median. Author's calculations.

Table 3. Housing in each country's own hours (local nominal GDP per hour worked), hours a year per 1,000 hours of housing capital, 16 countries (author's calculations). Columns 1–7: 1976–2020. Country rows: Table A1

	Total	Net rent	Valuation	Price ex val.	Needed	Net	Net ex val.	Net 1951–2020	Net 2021–2025
GDP-weighted	**85.9**	**44.0**	**4.1**	**37.8**	**46.3**	**39.6**	**35.5**	**35.9**	**46.2**
Median	96.7	42.3	9.0	39.0	53.2	47.5	29.2	40.1	35.5
Minimum	51.0	28.7	−6.7	17.8	25.9	21.5	4.1	18.9	4.3
Maximum	143.3	68.9	26.7	73.8	90.2	63.9	51.4	64.8	67.1

Measured in each country's own hours, housing gained a GDP-weighted 39.6 hours a year per 1,000 over 1976–2020 (35.5 excluding valuation). The country median was 47.5, from 21.5 in Japan to 63.9 in Norway. Every country is positive. The GDP-weighted figure sits below the median partly because Japan and Germany, with low housing returns, carry large weights. Over 1951–2020 the GDP-weighted net was 35.9 and the median 40.1; over 1976–2025, 40.3 and 47.5.

Rents grew more slowly than the hour. In national hours, price excluding valuation — roughly the growth of rents — was 37.8 hours a year, against 46.3 needed to keep pace with the hour. Housing's surplus over the hour is therefore its net rent yield, not rising prices. A home is a claim on a flow of housing services whose value, per unit, grew more slowly than the value of an hour of work.

4.3 Housing beside the other claims

Table 4. Net hour return of each claim, hours a year per 1,000 hours (author's calculations; same 16 countries and years; Macrohistory Database total returns). National-hour columns average each country's net at each year's GDP weights, over the same country-years as Table 3. Bills stand in for deposits, which typically earn less

Claim	World, 1976–2020	National, GDP-w., 1976–2020	National, median, 1976–2020	National, GDP-w., 1951–2020	National, median, 1951–2020
Housing	40.0	39.6	47.5	35.9	40.1
Equities	50.4	44.4	51.5	36.0	34.2
GMP (note 3)	38.0	–	–	–	–
Government bonds	27.9	26.7	27.8	0.9	−1.2
Bills	−1.6	−1.2	−4.7	−12.1	−19.5

In world hours over 1976–2020, equities gained 50.4 hours a year per 1,000, housing 40.0, the GMP 38.0, government bonds 27.9 and bills –1.6. Over the longer 1951–2020 span in national hours, bills lost ground to the hour in most countries, consistent with post-war inflation and repressed rates: a deposit shrank in hours. Government bonds roughly kept pace on GDP-weighted average (0.9) but lost ground in the median country (–1.2). Housing and equities gained.

4.4 Rent yields, and a correction to note 3

Table 5. Housing net rent yield and the GMP's cash distribution yield, % (author's calculations). Rent yield: 16 countries, weighted by same-year US-dollar GDP; 2025 chained with OECD price and rent indices

	1950	1976	1981	1990	2000	2010	2020	2025
Net rent yield, corrected weights	4.35	4.61	4.32	4.16	4.47	4.30	3.72	3.26
GMP cash distribution yield (note 3)	–	7.15	10.18	6.69	4.48	3.60	1.94	3.23
Note 3's calculation as reproduced here (wrong weights; printed as 5.5, 4.3, 2.7)	–	*5.52*	–	–	*4.28*	–	*2.73*	–

Correction to note 3, Section 7. Note 3 (Nakanishi 2026d, v1.4) reported the net rental yield on housing, weighted by GDP in US dollars across 16 countries, as 5.5% in 1976, 4.3% in 2000 and 2.7% in 2020 (5.52, 4.28 and 2.73 in this note's reproduction of note 3's calculation). Those weights were wrong. The Macrohistory Database records nominal GDP in millions, billions or trillions of local currency, depending on the country, and note 3 divided it by the exchange rate as if it were all in the same unit. As a result, Japan, the United States, Germany, France, the United Kingdom and Italy together carried an average weight of 0.6% over 1976–2020, while Spain carried 22%. With each country's GDP rescaled to US dollars (Section 3), the yields are **4.61%, 4.47% and 3.72%**. The substantive change is in the trend. Note 3's figures had the rent yield halving between 1976 and 2020, in the same direction as the GMP's cash yield; note 3 said only that "the direction is comparable". Corrected, it fell by about a fifth (4.61% to 3.72%, –19%) while the GMP's cash yield fell by almost three-quarters (7.15% to 1.94%, –73%). In 2020 the rent yield was above the GMP's cash yield in note 3's figures too; corrected, the margin is 1.78 points rather than 0.79. No other figure in note 3 used these weights. The error has been corrected in note 3 v1.5 (Nakanishi 2026d, v1.5). The independent check reproduced both the printed and the corrected values.

From 1976 to 1999 the GMP's cash yield was above the rent yield in 92% of years. From 2010 to 2020 the rent yield (between 3.7 and 4.5%) was above the GMP's cash yield (between 1.9 and 3.6%). In 2025 the two were about equal, 3.26% and 3.23%. The rent yield moved much less than the GMP's yield, which is dominated by bond coupons that reset with interest rates.

Neither side is net of tax. Following Jordà et al. (2019), a property tax of one point a year lowers housing's 1976–2020 net from 40.0 to 30.0 hours. Transaction costs of 7.7% of value per round trip cost 8.0 hours a year over a ten-year hold and 4.0 over a twenty-year hold; with both, the net is 22.0 (22.6 over 1976–2025). The GMP's figures carry no fees or withholding taxes either, but those are much smaller. Before taxes and costs, then, the home earned about as much as the world's investable capital; after a property tax and transaction costs it earned less: 22.0 hours against 38.0.

5. The household balance sheet in hours

5.1 What households hold

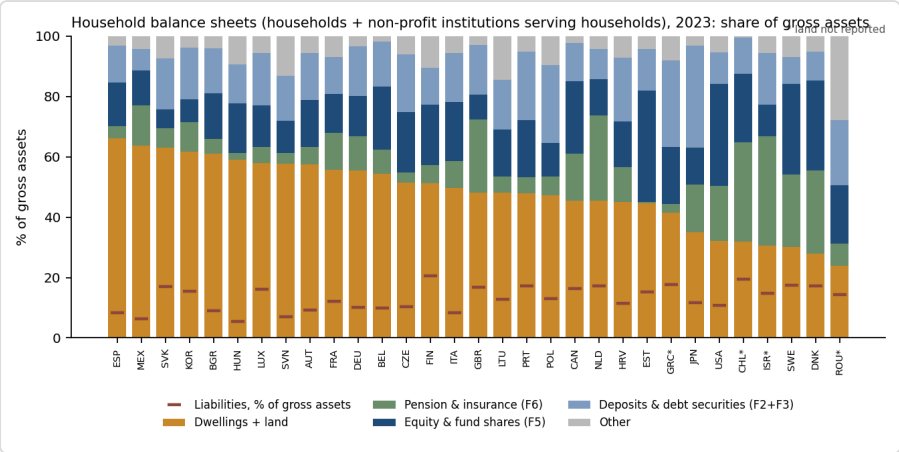


Figure 3. Household balance sheets (households and non-profit institutions serving households), 2023: shares of gross assets, and liabilities as a % of gross assets (marks). * land not reported. Author's calculations from OECD Tables 720 and 9B.

Table 6. Household balance sheets (households and non-profit institutions serving them), 2023, 31 countries: shares of gross assets, % (author's calculations from OECD Tables 720 and 9B). Years of work = net worth ÷ GDP. Country rows: Table A2

	Dwellings + land	Equity & fund shares (F5)	Pension & insurance (F6)	Deposits & debt sec. (F2+F3)	Other	Liabilities	Net worth, years of work	Net worth per worker, world hours
GDP-weighted	**41.6**	**23.3**	**16.0**	**14.2**	**4.9**	**11.7**	**–**	**–**
Median	48.2	15.6	8.1	15.5	5.5	13.0	3.6	17,092
Minimum	23.9	6.4	0.5	7.2	0.6	5.5	1.1	2,659
Maximum	66.2	36.8	36.2	33.8	27.9	20.6	6.0	52,361
United States	32.1	33.7	18.3	10.5	5.3	10.8	6.0	52,361
Japan	35.1	12.1	15.8	33.8	3.2	11.7	5.1	16,979

Weighted by GDP, dwellings and land were 41.6% of gross household assets in 2023 (median 48.2%). Equity and fund shares were 23.3% (median 15.6%): listed shares 8.7, unlisted 1.8, other equity 5.6 and investment-fund shares 6.3 (the sub-items sum to 22.4 because four countries do not report all of them). Pension and insurance claims were 16.0% (median 8.1%), deposits and debt securities 14.2% (median 15.5%), and other assets 4.9%. Liabilities were 11.7% of gross assets.

The range is wide. Dwellings and land run from 66% in Spain to 28% in Denmark (24% in Romania, which reports no land). Japanese households hold 34% in deposits and debt securities. US, Swedish and Danish households hold about 34, 30 and 30% in equity and fund shares. Countries that publish financial accounts but no dwellings (among them Switzerland, Norway, Ireland and New Zealand), and Australia, which publishes the reverse, are left out.

In hours of work. The median household net worth per worker was 3.6 years of the country's own work in 2023, or 5,851 hours. It was 6.0 years in the United States (OECD basis; Z.1 gives 5.60), 5.4 in France, 5.1 in Japan, 4.6 in Germany and 4.0 in the United Kingdom. In a balanced panel of 26 countries, the GDP-weighted figure rose from 3.49 years in 1995 to 6.00 in 2021 and fell back to 5.32 in 2023 (median 3.08, 3.81 and 3.62). In world hours the median was 17,092 hours per worker: 52,361 in the United States, 16,979 in Japan. Japan's figure was 27,961 in 2007; it fell because of the yen, while its figure in Japanese hours rose.

5.2 The United States since 1952

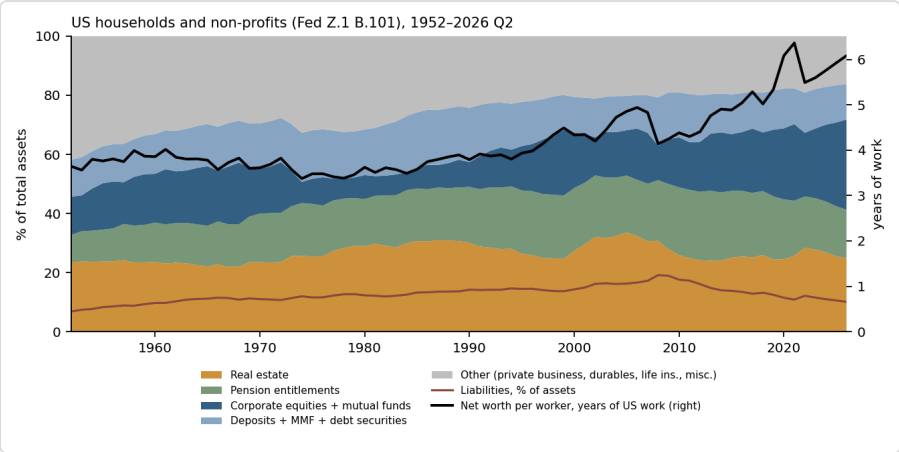


Figure 4. US households and non-profits (Federal Reserve Z.1, B.101), 1952–2026 Q2: shares of total assets (areas), liabilities (line), and net worth per worker in years of US work (right axis). Author's calculations.

Table 7. US households and non-profits, Federal Reserve Z.1 table B.101, % of total assets at year-end (2026: 2026 Q2; its world-hour value uses the world hour of March 2026, the last available) (author's calculations). On the OECD basis of Table 6, US net worth per worker is 6.0 years in 2023, against 5.60 here

	1952	1970	1990	2000	2007	2019	2026
Real estate	23.3	23.5	30.1	27.3	30.6	24.4	24.8
Pension entitlements	9.4	16.4	18.9	21.3	19.5	21.4	16.4
Corporate equities + mutual funds	12.9	15.2	8.4	18.8	17.2	22.5	30.6
Deposits, money-market funds, debt securities	12.6	15.3	18.4	12.0	12.7	13.3	12.0
Other (private business, durables, life insurance)	41.8	29.5	24.2	20.6	20.0	18.4	16.2
Liabilities, % of assets	6.8	11.0	14.2	14.3	17.2	12.5	10.0
Net worth per worker, years of US work	**3.64**	**3.61**	**3.79**	**4.34**	**4.83**	**5.34**	**6.09**
… in US hours	7,506	6,974	6,962	8,186	8,834	9,591	10,890
… in world hours	–	45,047	30,725	42,319	37,800	43,391	54,541

From 1952 to 2026, real estate moved from 23% to 25% of US household assets. Pension entitlements rose from 9% to 16% (peak 24% in 2013). Equities and mutual funds rose from 13% to 31%. "Other" fell from 42% to 16%, mostly because private businesses fell from 23% to 8% of assets. Liabilities rose from 7% to 10% of assets, with a peak of 19% in 2008.

Net worth per worker went from 3.64 years of US work in 1952 to 5.60 in 2023 and 6.09 in 2026. Over 1952–2023 it grew 0.60% a year in years of work but only 0.41% in US hours (from 7,506 to 10,013), because annual hours per worker fell from 2,060 to 1,789 (0.20% a year). In world hours it went from 45,047 in 1970 to 54,541 in 2026 (Z.1 basis throughout; the OECD basis of Table 6 gives 6.0 years in 2023, see the note to Table 7).

5.3 The claim the balance sheet leaves out

Table 8. Accrued-to-date pension entitlements, 2021 (Eurostat national accounts Table 29; author's calculations). Social-security (pay-as-you-go) schemes S13PS and funded schemes S1PF; PAYG % of GDP on Eurostat's own GDP. Net worth is households and non-profits (S1M). Country rows: Table A3

	PAYG, % of GDP	Funded, % of GDP	PAYG, % of household net worth	Net worth, years of work (= years of GDP)	… including PAYG
Median	**270**	**11**	**78**	**3.5**	**6.5**
Minimum	2	0	0	1.3	4.0
Maximum	452	203	300	6.2	10.7
Germany	273	27	59	4.7	7.4
France	345	0	58	5.9	9.4
Italy	430	9	73	5.9	10.2
Netherlands	209	203	38	5.6	7.6

Social-security pension promises are not in the standard household accounts. In 2021, across 30 countries with Table 29 data, their accrued value was a median 270% of GDP, from 2% in Denmark to 452% in Spain. Funded schemes were a median 11%. Where household net worth is known (24 countries), the pay-as-you-go claim was a median 78% of it: 44% in Belgium, 59% in Germany, 58% in France, 73% in Italy, 73% in Spain, 106% in Finland, 161% in Poland and 300% in Romania. Adding it raises median household wealth from 3.47 to 6.54 years of GDP. In most of Europe, then, the standard balance sheet leaves out close to half of household wealth once the pension claim is counted. These are present values under national assumptions; France's 2021 value is below its 2018 value, partly because of a change in discount rate.

5.4 What the balance sheet earns: a weighted average of index returns

Table 9. A weighted average of index returns, not a measured household return: 2023 balance-sheet shares (%) times each claim's net return in national hours, 1976–2020, hours a year per 1,000 (author's calculations). Last column: median of the 10 countries with Eurostat data; the same countries have a median of 42.4 without PAYG

Country	Dwellings + land	F5	F6	F2+F3	Other	Liab.	Housing alone	Gross assets	Net worth	PAYG (hours growth)	Net worth incl. PAYG
BEL	54	21	8	15	2	10	55.6	45.8	49.2	1.1	34.6
DEU	55	13	11	16	3	10	27.6	23.8	25.9	−2.9	15.2
DNK	28	30	28	10	5	17	33.9	50.2	57.0	0.2	56.7
ESP	66	14	4	12	3	8	26.4	20.7	22.1	2.0	13.6
FIN	51	20	6	12	10	21	57.1	43.3	53.2	−1.0	25.2
FRA	56	13	12	12	7	12	39.9	32.2	35.5	−0.4	22.3
GBR	48	8	24	16	3	17	36.2	29.4	34.6	1.7	–
ITA	50	20	9	16	6	8	26.2	20.2	21.1	0.0	12.2
JPN	35	12	16	34	3	12	21.5	12.7	13.8	−2.2	–
NLD	45	12	28	10	4	17	59.2	52.0	59.9	7.6	45.6
PRT	48	19	5	23	5	17	53.1	12.2	16.6	7.6	12.7
SWE	30	30	24	9	7	18	61.8	57.1	68.9	4.6	51.2
USA	32	34	18	10	5	11	49.9	46.6	50.8	11.1	–
Median							**39.9**	**32.2**	**35.5**	**1.1**	**23.8**

Table 9 multiplies each country's 2023 balance-sheet shares by the net return of the matching index in national hours over 1976–2020 (Section 4). Dwellings and land take the Macrohistory housing return; equity and fund shares the equity return; deposits the bill rate; debt securities the government bond return; pension and insurance claims 60% equities

and 40% bonds; other assets the bill rate; liabilities the bill rate plus 10 hours per 1,000. This is index arithmetic, not a measured household return: there are no taxes, costs, selection or leverage spreads beyond that one point.

Across 13 countries the median is 35.5 hours a year per 1,000 of net worth, GDP-weighted 41.3 (United States 50.8). The range runs from 13.8 in Japan to 68.9 in Sweden. Portugal's 16.6 reflects its equity return (−38.1) despite housing at 53.1. The GMP's net over 1976–2020 was 38.0 in world hours (Table 2), but the units differ: national hours here, world hours there. Average rather than 2023 shares give a median of 36.4; other assets at zero, 35.6; liabilities at the bill rate, 36.9.

Adding the pay-as-you-go claim at its wage-bill return (Section 6) lowers the European figures. Across the 10 countries with both, the median falls from 42.4 to 23.8 hours a year per 1,000: Germany from 25.9 to 15.2, Finland from 53.2 to 25.2.

6. The pension in hours

6.1 Why a claim on the wage bill earns the growth of hours worked

Samuelson (1958) and Aaron (1966) showed that a pay-as-you-go pension can pay each generation a return equal to the growth of the wage bill: the growth of pay per hour plus the growth of hours worked. Measure that claim in national hours, i.e. divide by GDP per hour, and the first part cancels if pay per hour grows with GDP per hour. What remains is the growth of total hours worked. If the labour share falls, pay per hour grows more slowly than GDP per hour, and the claim earns less than hours growth. This is the return the system can sustain, not the indexation rule of any one country; notional defined-contribution schemes credit accounts with wage-bill, average-income or GDP growth, depending on the country. Individual entitlements are usually revalued with average wages or with prices. A claim revalued with average pay per worker earns, in national hours, the growth of hours per worker, which was negative (−3.6 per 1,000 a year over 1976–2020, Table 10). Over 1976–2019 the labour share of the 16 Macrohistory countries, weighted by purchasing-power GDP, fell from 64.3% to 59.8% (Penn World Table); a wage-bill claim would then have earned about 1.7 hours per 1,000 a year less than hours growth (1.8 with US-dollar weights) (Karabarbounis and Neiman 2014 document the fall). The identity holds in the stylised lives of Section 7 exactly: a 10% contribution builds a claim worth, in years of final pay, 10% times the sum of each year's ratio of total hours at 65 to total hours in that year.

6.2 The growth of hours worked

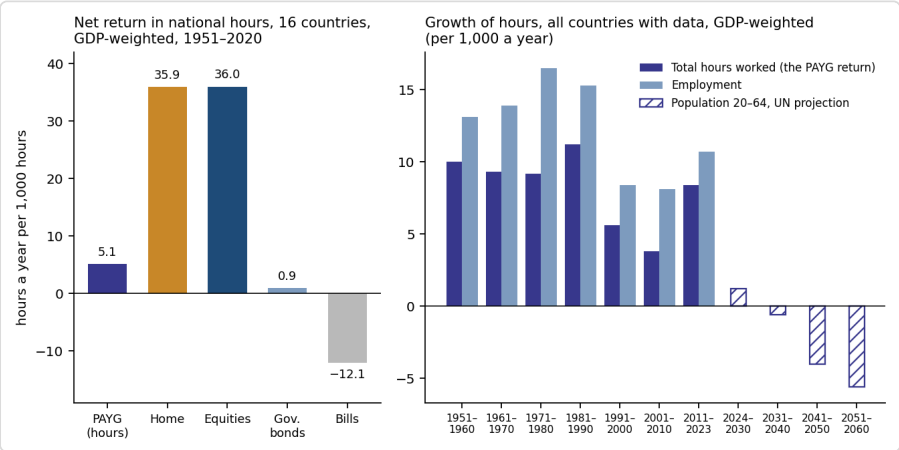


Figure 5. Left: net return of each claim in national hours, 16 Macrohistory countries, GDP-weighted, 1951–2020; the PAYG return is the growth of total hours worked. Right: growth of total hours worked and of employment by decade, all countries with Penn World Table hours, GDP-weighted; hatched bars are the UN projection of the population aged 20–64, at 2023 GDP weights. Author's calculations.

Table 10. Growth of total hours worked, GDP-weighted across countries, per 1,000 a year (log growth in % × 10) (author's calculations from Penn World Table 11.0). Employment and hours per worker do not add exactly to total hours because the country sets with each series differ slightly

	1951–1960	1961–1970	1971–1980	1981–1990	1991–2000	2001–2010	2011–2020	2021–2023	2011–2023	1951–2023	1976–2020
Total hours worked: the PAYG return	**10.0**	**9.3**	**9.2**	**11.2**	**5.6**	**3.8**	**2.2**	**28.8**	**8.4**	**8.2**	**6.6**
— employment	13.1	13.9	16.5	15.3	8.4	8.1	8.1	19.2	10.7	12.2	11.0
— annual hours per worker (return of a claim on average pay)	−2.7	−4.3	−5.8	−2.9	−2.3	−3.7	−5.8	9.7	−2.2	−3.4	−3.6
Total hours, constant 1950 sample	8.8	8.9	7.7	9.7	4.6	0.6	2.5	31.4	9.2	7.2	5.3
Total hours, 16 Macrohistory countries	8.4	8.1	4.9	8.1	3.9	−1.1	3.3	27.1	8.8	6.0	4.4
Memo: population 20–64, same countries	8.4	12.4	14.5	13.2	9.3	9.7	5.9	2.3	5.1	10.1	10.0
Countries (average)	28	31	44	48	61	98	129	128	129	66	80

Weighted by GDP across all countries with data, total hours grew 8.2 per 1,000 a year in 1951–2023. Employment grew 12.2; hours per worker fell by 3.4 per 1,000 a year in the countries with that series (the two do not add exactly to total hours because the country sets differ). By decade the growth of total hours was 10.0, 9.3, 9.2 and 11.2 to 1990, then 5.6 and 3.8, and 2.2 in 2011–2020, then 28.8 in 2021–2023 as hours recovered from 2020. The country set grows over time; a constant 1950 sample gives 7.2 for 1951–2023.

Table 11. The PAYG return by country: growth of total hours worked, per 1,000 a year (author's calculations; Penn World Table 11.0; UN World Population Prospects 2024, medium variant, for the projection)

Country	Hours, 1951–2023	Hours, 1976–2020	Employment, 1976–2020	Hours per worker, 1976–2020	Hours, 2001–2023	Population 20–64, 2024–2060 (projection)
Australia	15.9	15.2	17.5	−2.2	14.4	4.8
Belgium	1.3	1.1	5.8	−4.7	9.2	−2.6
Switzerland	7.4	5.7	10.9	−5.2	7.3	−3.9
Germany	−2.6	−2.9	4.2	−7.1	1.9	−7.0
Denmark	1.2	0.2	4.3	−4.2	3.9	−1.3
Spain	4.0	2.0	7.4	−5.4	6.8	−9.7
Finland	−0.6	−1.0	2.4	−3.4	3.9	−3.7
France	−0.2	−0.4	6.1	−6.5	5.5	−0.9
United Kingdom	1.2	1.7	5.9	−4.2	7.5	1.0
Italy	1.3	0.0	4.8	−4.8	2.8	−11.3
Japan	5.1	−2.2	3.0	−5.2	−3.1	−10.0
Netherlands	8.4	7.6	11.6	−4.0	8.7	−1.6
Norway	3.8	6.4	10.8	−4.4	9.7	−1.5
Portugal	5.9	7.6	6.4	1.1	−0.4	−6.9
Sweden	3.3	4.6	4.6	−0.1	8.0	0.2
United States	11.8	11.1	11.9	−0.9	5.8	1.6

Over 1976–2020, total hours grew 6.6 per 1,000 a year across all countries, 4.4 across the 16 Macrohistory countries, and a median 1.1 across the 13 countries of Table 9. Hours fell in Germany (−2.9), Japan (−2.2) and Finland (−1.0) and were about flat in France (−0.4) and Italy (0.0). In the United States they grew 11.1.

6.3 The pension beside the home and the portfolio

The like-for-like comparison is in national hours on the same countries. Across the 16 Macrohistory countries over 1951–2020, GDP-weighted, the wage-bill claim earned 5.1 hours a year per 1,000, against 35.9 for housing, 36.0 for equities, 0.9 for government bonds and −12.1 for bills (Figure 5, left). Over 1976–2020 it was 4.4 against 39.6 for housing. In world hours over 1976–2020, the growth of hours across all countries (6.6) can be set beside housing (40.0) and the GMP (38.0) only in aggregate, and the country sets differ. Over the longer span the wage-bill claim beat bills, which lost to inflation, and bonds, which roughly kept pace, and fell far short of the home and equities.

6.4 Projection

The UN medium-variant projection of the population aged 20–64, weighted by 2023 GDP, grows 1.2 per 1,000 a year in 2024–2030; its growth is then −0.6 in 2031–2040, −4.0 in 2041–2050 and −5.6 in 2051–2060. The world's headcount aged 20–64 still grows (5.0 over 2024–2060), because growth moves to countries with small GDP weights. Across the 16 Macrohistory countries it shrinks by 1.4 per 1,000 a year over 2024–2060. Over 1976–2020 the population aged 20–64 across all countries grew 13.9 per 1,000 a year at 2023 GDP weights (10.0 at each year's weights, Table 10). This is a projection, not data, and it is a population, not hours worked: participation and working time can move either way. But a claim on the wage bill, whose return in hours is the growth of hours worked, faces on these projections a negative return in most of the countries where such claims are large.

7. A life

7.1 Design

The life-cycle exercise follows 476 stylised workers: one for each of the 16 Macrohistory countries and each cohort that starts work at 25 in a year from 1950 to 1980 and retires at 65, between 1990 and 2020 (496 possible lives; Germany before 1961 and Japan before 1959 lack the data; Table A1's long-period column also needs national hours and the change in rent yield, so it starts later). Time steps are annual and flows come at the end of each year.

- **Pay** is the national hour with hours per worker held fixed; before 1960 it is chained back with the database's GDP per head. It is not measured earnings: where hours per worker fell, as in France, Japan and Sweden, actual annual pay grew more slowly. Because pay here is output per hour at fixed hours, not earnings, a home at 4 times pay or a 35% budget is a larger multiple of actual earnings than it looks: at the average labour share of 62% (1951–2019, Penn World Table), about 6.5 years and 57% of earnings.
- **Budget.** Each year 35% of pay goes to housing and saving; the rest is consumed and is the same for every rule.
- **The home** costs 4 times pay in the cohort's first full year of pay (the year after it starts work at 25, age 26), and its price then follows the database's nominal house price index. Gross rent is the net rent yield plus running costs of 2.0% of the price.
- **R (rent and invest)** pays gross rent and saves the rest of the budget in a domestic 60/40 portfolio of the database's equity and bond returns. In the GMP variant the account holds the domestic 60/40 until 1975 and the GMP, in US dollars converted to local currency at the database's year-end exchange rate, from 1976.
- **O (own)** rents and saves as R until 35, then buys if the account holds a 20% deposit plus 3.85% buying costs (otherwise it waits). The mortgage is 80% of the price, a 25-year annuity at the long rate plus 1.0 point, fixed at purchase. O then pays the mortgage and running costs and saves the rest. Wealth at 65 is the home less any remaining loan plus the account.
- **Overdraft.** Under both R and O the account may go below zero; a negative balance is charged the bill rate plus 1.0 point.
- **The pension.** On top of the budget, 10% of pay goes either into a pay-as-you-go claim whose value grows with the national wage bill (nominal GDP, i.e. the wage bill at a constant labour share), into a claim revalued with average pay per worker, or into the same 60/40 account.
- **Valuation.** O is re-run with the rent–price ratio frozen at its purchase-year level; the difference is the part of O's wealth that came from valuation.

Wealth is reported in years of own final pay. The model checks itself: with all returns set to zero, R's wealth equals the sum of its contributions; the funded claim equals 10% of summed pay; the pay-as-you-go claim equals the hours-growth formula of Section 6.1; and every decomposition adds up to machine precision.

Neighbouring cohorts share 39 of their 40 years, so the 476 lives contain far fewer independent histories than their number suggests: in effect 16 countries and a few distinct periods. Shares of lives are descriptions of this history, not probabilities.

The design was corrected once, after the first results were seen. The design fixed before any result priced the same standard home for every cohort at 4 years of pay in 2000, and did not let the account go below zero. That home cost far more than 4 years of pay in the 1950s (Table 15), and most early renters could not pay the rent from their first working year. The baseline was therefore changed to a home costing 4 years of pay in each cohort's first working year, and negative balances were allowed at the bill rate plus one point. The original design is kept as a robustness row (Table 16): its median wealth at 65 is 3.8 (R) and 4.2 (O) years of final pay, 303 renters go negative, and 158 owners never buy. Because the baseline was chosen with the results in view, Section 7 is the least protected part of this note against the author's choices.

7.2 Results

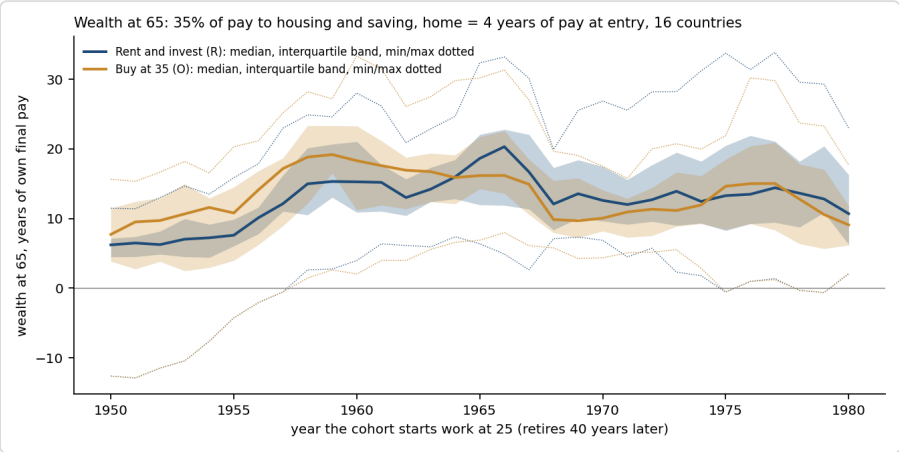


Figure 6. Wealth at 65 in years of own final pay, renting and investing (R) against buying at 35 (O), by the year the cohort starts work: median across 16 countries, interquartile band, and minimum and maximum (dotted). Author's calculations.

Table 12. Wealth at 65 in years of own final pay, 476 lives (16 countries × cohorts starting work 1950–1980) (author's calculations). Last column: the 300 lives whose account never went below zero

	Median	Interquartile range	Min to max	GDP-weighted mean	Median, never negative
R: rent and invest	12.06	8.04–18.10	−12.87 to 33.84	12.64	14.05
O: buy at 35	12.79	7.92–17.70	−12.87 to 33.29	12.84	15.59
O, net of selling cost	12.64	7.85–17.57	−12.87 to 33.14	12.73	15.51
O − R	**0.00**	**−2.95–3.34**	**−17.18 to 11.52**	**0.20**	**1.52**
O ÷ R	**1.00**	**0.78–1.29**	**0.19 to 3.45**	**1.02**	**1.10**
PAYG claim indexed to the wage bill, 10% of pay	4.28	4.00–4.73	3.22 to 5.96	4.61	4.35
Funded claim, 10% of pay in the 60/40	11.09	7.86–14.25	1.98 to 28.43	10.67	10.73
Funded ÷ PAYG	**2.44**	**1.86–3.17**	**0.38 to 7.56**	**2.31**	**2.41**
Share of O's wealth from valuation, %	2.1	−1.5–8.2	−73.8 to 86.6	1.4	1.7
Buying age	35	35–40	35 to 64	37	35
R in world hours ÷ fixed annual hours per worker	57.1	32.7–82.0	−73.6 to 211.1	56.2	62.8
O in world hours ÷ fixed annual hours per worker	57.2	31.9–84.2	−73.6 to 177.1	57.6	66.2

The median renter ends with 12.06 years of final pay and the median owner with 12.79 (12.64 after selling costs). The median ratio of O to R is 1.00, with an interquartile range of 0.78–1.29; the median difference is 0.00 years of final pay

(interquartile range −2.95 to 3.34; mean −0.08). The median ratio of exactly 1.00 and difference of exactly 0.00 arise because the 17 lives that never buy have O = R and sit at the middle of the distribution. Among the 459 buyers the median difference is 0.08 (interquartile range −3.10 to 3.48; Table 13), and the owner wins in 51%. The difference is the safer headline: where the renter's wealth is near or below zero the ratio loses its meaning. The owner comes out ahead in 236 of 476 lives (50%). Among the 300 lives whose account never went below zero, the median ratio is 1.10, the median difference 1.52 and the owner wins in 59%. The median buyer buys at 35; 17 lives never buy. In world hours, one hour of own final pay buys a median 4.6 world hours; wealth in world hours divided by the fixed annual hours per worker is a median 57.1 for R and 57.2 for O.

Table 13. Why owning and renting come out close: O − R for the 459 lives that buy, years of own final pay (author's calculations). Costs are shown as positive numbers to be subtracted. Means add up exactly; medians do not

Item	Median	25th pct.	75th pct.	Mean	Median, never negative
Home price gain	2.20	1.43	3.49	2.50	2.38
Imputed rent saved (gross)	3.16	2.26	4.12	3.23	3.34
Running costs paid by the owner (−)	0.94	0.65	1.41	1.05	1.04
Net rent saved	2.13	1.55	2.70	2.17	2.17
Mortgage interest (−)	1.10	0.49	1.79	1.20	0.95
Buying cost (−)	0.03	0.02	0.05	0.03	0.02
Forgone 60/40 return on the deposit and extra outlays, and the difference in overdraft interest (−)	2.76	0.15	6.53	3.52	2.15
O − R	**0.08**	**−3.10**	**3.48**	**−0.08**	**1.52**

Why the two come out close. The means add up exactly. On average across the 459 buyers, the home's price gain (2.50 years of final pay) plus the rent saved net of running costs (2.17), less mortgage interest (1.20), buying costs (0.03) and the 60/40 return forgone on the deposit and the extra outlays (3.52), gives −0.08: 2.50 + 2.17 − 1.20 − 0.03 − 3.52 = −0.08. The last term also contains the difference in overdraft interest between O and R. The medians in Table 13 show the typical size of each item but do not add up. A home, bought with a mortgage, is a leveraged claim on housing; renting and investing is a claim on the 60/40. Over the 1950–2020 working lives their returns were close, so the result turns on the cost of the debt and on how well the 60/40 did.

By cohort. The median ratio of O to R was 1.27 for cohorts starting in the 1950s, 1.02 for the 1960s, 0.87 for the 1970s and 0.76 for the 1980 cohort. Table 14 measures where the decline comes from. Two changes each remove most of it: variable-rate mortgages remove 56% (the ratio falls only from 1.17 to 0.95), and bills as the saving vehicle remove all of it (the ratio stays between 1.43 and 1.50). The decline therefore needs both: fixed-rate debt, which inflation eroded for early buyers and which locked later buyers into high rates, and the comparison with a strong 60/40 portfolio after 1980, which later renters held for most of their working lives. The two overlap and cannot be added. The US 1970 cohort is an example: it bought in 1982, at 37, at a mortgage rate of 14.0%, paid 2.34 years of final pay in interest, overdrew to −0.52 years of pay, and ended with 4.75 against the renter's 9.60.

Table 14. Owning against renting by decade of the cohort's start, under three variants: medians of O ÷ R and O − R in years of own final pay (author's calculations). "1980s" is the 1980 cohort only

Cohort	Baseline: O ÷ R	O − R	Variable-rate mortgage: O ÷ R	O − R	Bills vehicle: O ÷ R	O − R
1950s	1.27	2.67	1.17	2.03	1.43	2.56
1960s	1.02	0.28	1.04	0.41	1.50	3.08
1970s	0.87	−1.47	0.98	−0.26	1.49	2.41
1980s	0.76	−3.32	0.95	−0.91	1.46	1.58

Valuation is a small part. Holding the rent–price ratio at its purchase-year level, valuation accounts for a median 2.1% of the owner's wealth at 65 (interquartile range −1.5 to 8.2%). The owner's gain over a life came from net rent and from rent growth, not from a rising price per unit of rent.

Lives that overdraw. In all, 176 lives go below zero at some point: 128 under R and 176 under O, and every R case is also an O case. The median lowest balance among them is −0.26 years of pay. By country (Table A4), the number of lives that overdraw under O is Finland 31, Norway 29, Sweden 24, the Netherlands 16, France 14, Australia 13, and 11 or

fewer elsewhere. In all, 12 lives end below zero; all of them never buy, so O and R are identical for them. The French cohorts of 1950–1957 reach as low as –12.9 years of pay: in the database, French house prices rose much faster than pay between 1951 and 1965, so a home worth 4 years of pay in 1951 was worth about 15 by 1965, with a net rent yield that put rent above a full year's pay. The others are cohorts in Finland (1950, 1978) and Norway (1975, 1979). In Finland, Norway and Sweden the database's rent yields are high relative to prices, so most lives there overdraw. The database's post-war rent levels in these countries may be imperfectly measured. An overdraft costs the bill rate plus one point, which is usually below the mortgage rate, so these lives borrow cheaply and unsecured; that flatters both R and O where overdrafts are large.

7.3 The pension over a life

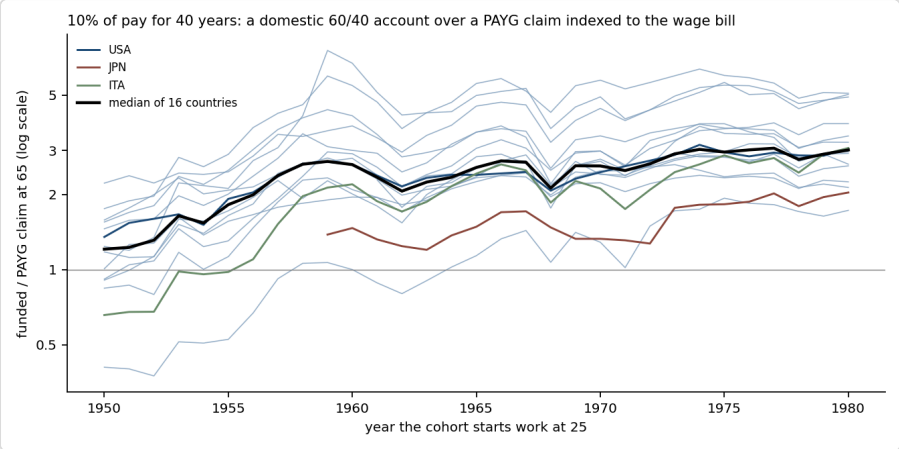


Figure 7. 10% of pay for 40 years: the funded claim (domestic 60/40) divided by the pay-as-you-go claim indexed to the wage bill at 65, by cohort; thin lines are countries. Log scale. Author's calculations.

A contribution of 10% of pay for 40 years builds a pay-as-you-go claim worth a median 4.28 years of final pay if it is indexed to the wage bill and 3.75 if it is indexed to average pay per worker, and a funded claim worth 11.09 (a median 2.95 times the average-pay claim). The average-pay figure depends on the framing: contributions here scale with pay at fixed hours while the claim is revalued with actual hours per worker; if contributions also fell with hours per worker, the claim would be exactly 10% × 40 = 4.00 years of final pay. The funded claim is a median 2.44 times the pay-as-you-go claim (interquartile range 1.86–3.17) and is larger in 95% of lives. That advantage is entirely the 60/40's return over the growth of the wage bill. With the GMP as the vehicle from 1976 the ratio is 1.88; with bills it is 0.91, and the funded claim wins in only 27% of lives. The pay-as-you-go claim is valued here at its indexation return, not at any country's real benefit rules, and there is no survival risk.

7.4 The price of a home in years of pay

Table 15. Price of a home that cost 4 years of pay in 2000, in years of pay (Macrohistory Database nominal house prices ÷ national GDP per hour; author's calculations)

	1950	1960	1970	1980	1990	2000	2010	2020
Median	**6.5**	**6.0**	**5.3**	**4.8**	**4.5**	**4.0**	**5.0**	**5.0**
25th percentile	4.9	4.7	4.6	3.4	3.8	4.0	4.0	4.4
75th percentile	8.4	8.7	6.2	5.2	5.5	4.0	5.8	6.0
Countries	15	15	16	16	16	16	16	16

The same home — an index of roughly constant quality — cost a median 6.5 years of pay in 1950, 5.3 in 1970, 4.0 in 2000 and 5.0 in 2020. From 1950 to 2020 the median price in years of pay fell by 0.36% a year, in line with Section 4's finding that rents grew more slowly than the national hour. The fall was not steady: price in years of pay rose by about a quarter between 2000 and 2020.

8. Robustness

Table A5 lists the housing variants. The tax and cost rows follow Jordà et al. (2019). Equal country weights raise housing's 1976–2020 net to 48.2; leaving out the United States lowers it to 35.3. Annual-average timing for the world hour gives 41.1. Weights are start-of-year; same-year weights give 42.6 with World Bank GDP at annual-average exchange rates, and 45.3 with the database's end-of-year rates, which build the year's currency move into its own weight. Interpolating the national hour to December-to-December timing gives a national median of 48.2 over 1976–2019, against 47.5 for the baseline on the same window. The national hour built from GDP per head instead of GDP per hour gives a median of 46.8; the two series' annual growth rates correlate at 0.90, with GDP per hour growing 0.24 points a year faster. OECD rent growth in place of the database's gives 41.7 (37.5 excluding valuation) and a national median of 45.8. None of these reverses the finding that, before taxes and costs, housing earned at least as much as the GMP; equal country weights (48.2 against 38.0) put housing clearly ahead.

Table 16. A life: one change at a time from the baseline, medians across lives (author's calculations). * Lives that never buy have O ÷ R = 1, so rows with many non-buyers show 1.00. PAYG: indexed to the wage bill

Variant	R	O	O ÷ R*	O wins, %	Val. share, %	Funded ÷ PAYG	Funded wins, %	O ever negative	Never buy
Baseline	**12.06**	**12.79**	**1.00**	**50**	**2.1**	**2.44**	**95**	**176**	**17**
Home 3× pay	19.00	19.64	1.04	58	0.8	2.44	95	37	6
Home 6× pay	0.88	0.94	1.00	21	5.0	2.44	95	415	239
Running costs 1.5%	14.09	15.14	1.03	53	2.1	2.44	95	141	12
Running costs 3%	8.81	8.26	0.96	39	2.4	2.44	95	258	40
Budget 25% of pay	2.08	2.23	1.00	29	7.8	2.44	95	391	182
Budget 45% of pay	23.30	24.39	1.04	57	0.9	2.44	95	40	6
GMP vehicle (from 1976)	9.33	10.88	1.16	63	3.0	1.88	97	176	19
Bills vehicle	5.19	8.02	1.48	80	4.9	0.91	27	200	22
Variable-rate mortgage	12.06	13.36	1.02	53	2.1	2.44	95	182	17
Home 4× pay in 2000 (original design)	3.77	4.23	1.00	30	11.6	2.44	95	346	158

For the life, the home's price, running costs and budget move the level of wealth a great deal, and the ratio of O to R little. The vehicle and the mortgage type matter more (Table 14): with the GMP, O beats R in 63% of lives (median ratio 1.16); with bills, in 80% (1.48). Across the full grid of 162 variants on the baseline anchor, the share of lives in which O beats R runs from 2% to 95% (median 58%), the median ratio from 0.95 to 1.53, and the median valuation share from –15% to 12%. Under the original 2000 anchor, O beats R in a median 30% of variants' lives and the valuation share is about 11%, larger mainly because buyers there buy later (median age 41) and more of them bought into the post-2000 rise. The funded-to-pay-as-you-go ratio does not depend on the housing choices at all.

9. What it means, and what it does not

What it means. In hours of work, the home and the investable portfolio were both claims on capital, and both earned, before taxes and costs, 3.5 to 4.1% a year more than the growth of the hour (1976–2025, world hours); after a property tax and transaction costs, the home earned less than the portfolio. They earned it differently: the home through rents that kept rising once valuation is set aside (price excluding valuation 39.6 hours a year against the GMP's 21.5), the portfolio with more help from falling yields (valuation 11.0 against 6.2). A household that owned its home and a household that rented and invested ended a working life in much the same place, and which one came out ahead depended on the cost of its mortgage and on how well the 60/40 did. The large difference is between those claims and a pension claim indexed to the wage bill. That claim keeps pace with work by design, and beyond that it earns only the growth of the amount of work done. That growth was under 1% a year, was negative in Germany, Japan and Finland and about zero in France and Italy over 1976–2020, and the UN projection of the population aged 20–64 turns negative on GDP-weighted average after 2030. A claim revalued with average pay per worker did worse still.

What it does not mean. It is not advice to buy, rent or opt out of a pension. A state pension also insures against living long, against inflation and against market losses, none of which is valued here. The comparisons are of indices:

- **An index is not a house.** There is no selection, no idiosyncratic risk, and no depreciation beyond the database's cost deduction. The 16 countries are survivors, all now rich; their housing and equity returns are likely higher than those of a random sample of countries over the same years.

- **No taxes.** No income, property, transfer or capital-gains tax, and no tax relief on mortgages or pensions, which in many countries favour both owning and funded saving. Imputed rent on an owner's home is untaxed in most countries, which favours owning.
- **Index arithmetic.** The balance-sheet return of Section 5.4 is a weighted average of index returns, not a measured household return.
- **Units.** The wage-bill return equals hours growth only if pay per hour keeps pace with GDP per hour; the falling labour share cost about 1.7 per 1,000 a year over 1976–2019 (Section 6.1). National and world hours answer different questions; Sections 4 and 6 compare like with like where they can.
- **Present values.** The Table 29 entitlements depend on national discount rates, which differ by country; no common-rate version is available.
- **The life design** was corrected after the first results were seen (Section 7.1); the database's early rent levels in France and the Nordic countries drive the overdrafts; overdrafts are cheaper than mortgages.
- **Data ends.** The database ends in 2020; 2021–2025 uses OECD indices and rents of sitting tenants; the world hour after mid-2025 and the Penn World Table after 2023 are not yet data.

10. Disclaimer

This note is general information and a research record addressed to no particular reader. It does not recommend any security, fund, property, pension choice or transaction, does not suggest whether any person should buy or rent a home, and is not individualised advice. The author is not a regulated investment adviser or licensed financial services provider and does not respond to individual requests for advice. Historical and simulated figures do not indicate future results; the stylised lives are hypothetical and are not the results of any actual household. No representation is made as to the accuracy or completeness of the information, and the author accepts no liability for any loss arising from its use. All tables are the author's own calculations from public data; no figures from other publications are reproduced. Nothing here is legal, tax or pension advice.

The author resides in Japan. This note provides general information to an unspecified audience and does not constitute investment advisory business under the Financial Instruments and Exchange Act of Japan.

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Appendix A: country tables

Table A1. Housing in each country’s own hours (local nominal GDP per hour worked), hours a year per 1,000 hours of housing capital (author’s calculations). Columns 1–7: 1976–2020. Germany’s 1951–2020 column starts in 1963 and Japan’s in 1960

	Total	Net rent	Valuation	Price ex val.	Needed	Net	Net ex val.	Net 1951–2020	Net 2021–2025
Australia	103.7	36.7	19.7	47.4	58.7	45.0	25.4	40.3	35.8
Belgium	98.0	52.2	17.1	28.8	42.4	55.6	38.5	52.9	33.2
Switzerland	76.7	46.0	9.0	21.7	25.9	50.8	41.8	34.7	51.8
Germany	68.6	41.7	1.0	25.9	41.1	27.6	26.6	26.7	4.3
Denmark	85.0	39.8	8.3	36.9	51.0	33.9	25.7	41.4	43.3
Spain	98.8	28.7	22.2	47.9	72.5	26.4	4.1	20.2	35.2
Finland	115.0	68.9	9.0	37.1	58.0	57.1	48.1	64.8	24.1
France	91.3	39.1	15.3	36.8	51.4	39.9	24.6	56.3	15.2
United Kingdom	100.4	33.8	8.5	58.1	64.2	36.2	27.6	32.4	17.1
Italy	94.6	36.7	−3.9	61.8	68.4	26.2	30.1	18.9	39.5
Japan	51.0	40.0	−6.7	17.8	29.5	21.5	28.2	32.3	56.3
Netherlands	95.1	49.8	7.8	37.5	35.9	59.2	51.4	57.2	56.4
Norway	125.2	56.0	18.5	50.7	61.3	63.9	45.4	53.7	4.7
Portugal	143.3	42.9	26.7	73.8	90.2	53.1	26.5	35.4	67.1
Sweden	116.8	59.4	16.4	41.0	55.0	61.8	45.4	54.7	6.2
United States	95.3	50.6	4.4	40.4	45.4	49.9	45.6	40.0	62.7
GDP-weighted	**85.9**	**44.0**	**4.1**	**37.8**	**46.3**	**39.6**	**35.5**	**35.9**	**46.2**
Median	**96.7**	**42.3**	**9.0**	**39.0**	**53.2**	**47.5**	**29.2**	**40.1**	**35.5**

Table A2. Household balance sheets (households and non-profit institutions serving them), 2023: shares of gross assets, % (author’s calculations from OECD Tables 720 and 9B). Years of work = net worth ÷ GDP = net worth per worker in hours of national work ÷ annual hours per worker. * land not reported, so the housing share is understated

Country	Dwellings + land	Equity & fund shares (F5)	Pension & insurance (F6)	Deposits & debt sec. (F2+F3)	Other	Liabilities	Net worth, years of work	Net worth per worker, world hours
ESP	66.2	14.5	4.0	12.2	3.0	8.3	5.5	22,670
MEX	63.7	11.6	13.3	7.2	4.2	6.4	2.9	4,703
SVK	63.1	6.4	6.3	16.9	7.3	16.9	2.3	6,543
KOR	61.7	7.5	9.8	17.3	3.7	15.5	5.2	17,902
BGR	61.1	15.1	4.8	14.9	4.1	9.1	2.8	4,888
HUN	59.1	16.6	2.1	12.7	9.5	5.5	3.6	8,356
LUX	57.9	13.8	5.3	17.5	5.5	16.0	3.4	30,957
SVN	57.8	10.7	3.5	15.0	13.1	6.9	3.7	12,396

Country	Dwellings + land	Equity & fund shares (F5)	Pension & insurance (F6)	Deposits & debt sec. (F2+F3)	Other	Liabilities	Net worth, years of work	Net worth per worker, world hours
AUT	57.5	15.6	5.8	15.5	5.6	9.2	4.4	25,917
FRA	55.8	12.8	12.2	12.3	6.9	12.2	5.4	28,454
DEU	55.5	13.3	11.4	16.5	3.4	10.0	4.6	23,948
BEL	54.4	20.9	8.1	14.8	1.8	10.0	5.3	35,945
CZE	51.4	19.9	3.5	19.2	6.0	10.4	2.9	9,767
FIN	51.3	19.9	6.1	12.3	10.5	20.6	3.1	17,337
ITA	49.7	19.6	8.8	16.2	5.7	8.4	5.3	24,763
GBR	48.2	8.3	24.1	16.5	2.9	16.9	4.0	21,609
LTU	48.1	15.6	5.3	16.5	14.5	12.7	2.0	5,558
PRT	48.0	18.9	5.2	22.7	5.1	17.2	3.4	10,107
POL	47.2	11.2	6.2	25.9	9.6	13.0	1.6	4,078
CAN	45.4	24.0	15.7	12.6	2.2	16.4	5.3	30,285
NLD	45.4	12.2	28.2	9.9	4.3	17.2	4.8	28,117
HRV	45.1	15.3	11.4	20.9	7.2	11.5	2.5	6,404
EST	44.5	36.8	0.5	13.8	4.3	15.2	2.3	7,246
GRC*	41.5	18.9	2.9	28.6	8.1	17.7	2.4	6,175
JPN	35.1	12.1	15.8	33.8	3.2	11.7	5.1	16,979
USA	32.1	33.7	18.3	10.5	5.3	10.8	6.0	52,361
CHL*	32.0	22.6	32.9	11.9	0.6	19.5	2.0	4,140
ISR*	30.7	10.4	36.2	17.1	5.7	14.9	3.0	17,092
SWE	30.1	29.9	24.0	9.1	6.8	17.5	4.2	24,387
DNK	27.9	30.0	27.5	9.5	5.1	17.2	4.5	30,133
ROU*	23.9	19.4	7.3	21.6	27.9	14.2	1.1	2,659
GDP-weighted	**41.6**	**23.3**	**16.0**	**14.2**	**4.9**	**11.7**	**–**	**–**
Median	**48.2**	**15.6**	**8.1**	**15.5**	**5.5**	**13.0**	**3.6**	**17,092**

Table A3. Accrued-to-date pension entitlements, 2021 (Eurostat national accounts Table 29; author's calculations). Social-security (pay-as-you-go) schemes S13PS and funded schemes S1PF

Country	PAYG, % of GDP	Funded, % of GDP	PAYG, % of household net worth	Net worth, years of work (= years of GDP)	… including PAYG
ESP	452	5	73	6.2	10.7
ITA	430	9	73	5.9	10.2
GRC	403	2	159	2.5	6.6
SVN	389	9	110	3.6	7.4
AUT	382	12	81	4.7	8.5
ROU	378	7	300	1.3	5.0
FRA	345	0	58	5.9	9.4
FIN	333	4	106	3.1	6.5
HUN	322	0	98	3.4	6.7
HRV	319	31	118	2.7	5.9
EST	303	0	125	2.4	5.5
LUX	291	7	75	3.9	6.8
PRT	278	11	74	3.7	6.5
LTU	274	10	129	2.1	4.9
DEU	273	27	59	4.7	7.4
POL	268	8	161	1.7	4.4
CZE	261	9	84	3.0	5.5
BEL	259	27	44	5.9	8.5
CYP	251	15	–	–	–
NOR	242	41	–	–	–
CHE	235	145	–	–	–
LVA	230	20	136	1.7	4.0
NLD	209	203	38	5.6	7.6
MLT	208	0	–	–	–
BGR	196	14	65	3.0	5.0
SVK	194	11	75	2.6	4.5
SWE	187	110	38	5.0	6.9
IRL	105	27	–	–	–
ISL	97	201	–	–	–
DNK	2	70	0	5.0	5.0

Table A4. A life by country, medians across cohorts, years of own final pay (author's calculations)

Country	Lives	R	O	O ÷ R	O beats R, %	PAYG	Funded	Funded ÷ PAYG	Buying age	O ever negative
Australia	31	8.71	11.80	1.10	58	5.61	11.65	2.05	35	13
Belgium	31	11.04	14.38	1.52	94	4.14	12.00	2.93	38	1
Switzerland	31	13.77	15.69	1.10	81	4.56	11.36	2.46	35	2
Germany	20	20.16	16.01	0.80	0	3.81	10.05	2.60	35	1
Denmark	31	22.98	16.49	0.63	0	4.01	15.12	3.79	35	11
Spain	31	18.75	17.51	0.97	48	4.67	11.70	2.33	35	3
Finland	31	10.73	10.13	1.02	55	3.92	18.04	4.49	44	31
France	31	17.38	10.48	0.89	13	4.04	11.62	2.92	35	14
United Kingdom	31	13.28	14.79	1.04	58	4.19	9.91	2.34	35	5
Italy	31	16.99	14.16	0.96	48	4.13	8.86	2.11	35	6
Japan	22	8.33	7.29	0.90	9	3.71	5.61	1.48	35	1
Netherlands	31	11.76	10.84	1.07	55	4.74	16.43	3.43	37	16
Norway	31	7.51	8.82	1.34	74	4.41	10.01	2.29	42	29
Portugal	31	9.03	9.25	1.27	65	5.18	5.90	1.07	35	9
Sweden	31	16.68	15.25	1.02	52	4.24	22.91	5.09	40	24
United States	31	12.39	15.21	1.17	55	5.17	12.62	2.43	35	10

Table A5. Housing: robustness, net hours a year per 1,000 hours (author's calculations). World hours 1976–2020 unless stated. National-hour and GMP comparisons are in Tables 2, 3 and A1. "Dec-to-Dec interpolated hour": the national hour interpolated to December values instead of calendar-year averages. Tax and cost parameters from Jordà et al. (2019)

Case	Net	Net excluding valuation
Baseline: world hours, GDP weights	**40.0**	**35.9**
Property tax 1 pt a year	30.0	25.9
Transaction costs 7.7% round trip, 10-year hold	32.0	27.9
Transaction costs 7.7% round trip, 20-year hold	36.0	31.9
Property tax 1 pt + costs over 10 years	22.0	17.9
Equal weights	48.2	37.3
Ex-USA (GDP weights)	35.3	31.2
World hour on annual-average timing	41.1	36.9
National hours, Dec-to-Dec interpolated hour, median, 1976–2019	48.2	31.5
Baseline national hour, median, same window 1976–2019	47.5	29.8
National hours from GDP per head (Macrohistory Database), median	46.8	33.1
OECD rent growth for the database's implied rent growth, world, 1976–2020	41.7	37.5
OECD rent growth for the database's implied rent growth, world, 1976–2025	42.3	36.1
OECD rent growth, national-hour country median, 1976–2020	45.8	28.7
Equal weights, 1976–2025	46.2	34.5
Ex-USA, 1976–2025	33.0	27.6
Property tax 1 pt + 7.7% costs over 10 years, 1976–2025	22.6	16.4

Appendix B: reproduction

File	Content
`code/fetch_*.py`	Download and clean the public inputs (`data/SOURCES.md` lists URLs, dates and terms)
`code/build_housing.py`, `build_housing_ext.py`	Section 4: housing in world and national hours, 1951–2020 and the OECD extension to 2025; the note-3 correction; Tables 2–5, A1, A5; Figures 1–2
`code/build_balance.py`	Sections 5–6: balance sheets, Z.1, Eurostat Table 29, hours growth, projections; Tables 6–11, A2, A3; Figures 3–4
`code/build_life.py`	Section 7: the stylised lives, decompositions and robustness grid; Tables 12–16, A4; Figures 6–7
`code/build_extras.py`	Two measurement variants of the housing headline (portfolio-yield valuation, same-year weights); `output/housing/extras.json`
`code/build_paper.py`	All tables and every number in the text, generated from `output/`; Figure 5
`output/*/build_log.json`	Input checksums, identity checks, headline values
`check/`	Independent checks of Sections 4–7, written before reading the builders

Run: `code/fetch_*.py` → `build_housing.py` → `build_housing_ext.py` → `build_extras.py` → `build_balance.py` → `build_life.py` → `build_paper.py`. Inputs (58 files, SHA-256, recorded at run time in the build logs):

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```